

TruthScan Project Report

Project Overview

TruthScan is a comprehensive web-based platform designed for digital content verification and forensic analysis. The core objective of the project is to provide a robust suite of tools for detecting fake news, manipulated media (deepfakes), and file tampering, helping investigators and everyday users verify the authenticity of digital content.

Features & Implementation Progress

1. Backend Architecture (Flask & SQLite)

- **Framework:** Built a robust web application using Python and the Flask framework.
- **Database (`database.py`):** Designed an SQLite database to store user credentials (`users` table) and forensic analysis reports (`forensic_records` table) securely.
- **Routing:** Implemented structured routing for dashboards, history viewing, and individual forensic reports.

2. User Authentication System

- **Security:** Integrated `werkzeug.security` for cryptographic password hashing (`pbkdf2:sha256`).
- **Session Management:** Implemented secure session tracking to keep users logged in.
- **Access Control:** Built a custom `@login_required` decorator to restrict access to sensitive analysis tools and history logs, automatically redirecting unauthorized users to the login screen.

3. Machine Learning & Natural Language Processing

- **Fake News Detection (`ml_model.py`):** Developed an NLP pipeline utilizing `TfidfVectorizer` and `LogisticRegression` to classify text as "Potentially Fake" or "Potentially Real".
- **Source/Channel Verification Engine:** Built a `ChannelVerifier` module that checks source links (e.g., YouTube, Telegram, Twitter) against a pre-registered database of official channels (such as Defence Squad and government domains). The ML model adjusts its final confidence score based on the credibility of the source channel.

4. Forensic Analysis Modules (`utils.py`)

- **Image Analysis:** Calculates SHA-256 and Perceptual hashes to detect exact or visual duplicates, and extracts hidden EXIF metadata.
- **Video Analysis:** Processes video frames for deepfake detection signatures and metadata extraction.
- **File Integrity Checker:** Allows users to upload two files to perform a bit-by-bit cryptographic comparison (SHA-256) to verify if a file has been altered or tampered with.

5. Frontend & UI Redesign

- **Aesthetic Upgrade:** Transitioned to a mature, high-end minimalist light theme using a monochrome foundation (clean white `#F9F9FB`, deep charcoal `#111111`, and soft light-gray cards `#F1F1F4`).
- **Accent Details:** Applied an elegant slate-navy (`#3B4E68`) for interactive elements and buttons.
- **Dynamic Views:** Built Jinja2 templates (`base.html`, `login.html`, `register.html`, `report.html`) that dynamically update based on the user's session state and display verification badges (e.g., [Verified Official Channel]) seamlessly.

6. Deployment & Collaboration

- **Public Tunneling:** Configured `cloudflared` to expose the local Flask server to the public internet securely, generating live URLs (e.g., `trycloudflare.com`) for external access and testing.
- **Version Control:** Initialized a Git repository (`Keshav018-miet/TruthScan`), committed all features, and pushed them to GitHub, establishing a standard branching and pull-request workflow for multi-user

collaboration.

Conclusion

To date, TruthScan has evolved from a basic concept into a full-featured prototype with a secure authentication system, an intelligent ML-driven text analysis engine backed by source verification, cryptographic file integrity checking, and a polished, professional user interface ready for team collaboration.